

RESEARCH ARTICLE | SEPTEMBER 09 2026

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Rev. Sci. Instrum. 97, 094704 (2026)

<https://doi.org/10.1063/5.0324162>



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Cite as: Rev. Sci. Instrum. 97, 094704 (2026); doi: 10.1063/5.0324162

Submitted: 22 January 2026 • Accepted: 6 August 2026 •

Published Online: 9 September 2026



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ABSTRACT

Shielding effectiveness is the core metric characterizing a material's electromagnetic shielding performance. With respect to measuring shielding effectiveness for thin materials over an ultra-wide frequency range, existing methods suffer from an inherent bottleneck: they cannot achieve broadband testing that simultaneously accommodates low-frequency cutoffs and high-frequency upper limits. To address the bottleneck where existing methods struggle to cover ultra-wide frequency bands starting from 10 MHz, this paper effectively overcomes measurement limitations within the 10 MHz–8 GHz ultra-wideband range by specifically optimizing the support medium and flange structures of the coaxial flange assembly. This establishes a comprehensive electromagnetic shielding effectiveness measurement methodology suitable for ultra-wide frequency domains. To verify the effectiveness and reliability of the proposed method, this study selected carbon nanotube materials as representative test samples for comparative validation experiments. The results show that the deviation between the values measured by this method and the standard reference values is less than 0.5%. A quantitative assessment performed in accordance with the Guide to the Expression of Uncertainty in Measurement confirms that the measurement accuracy of this method fully meets the electromagnetic shielding effectiveness testing requirements specified by the American Society for Testing and Materials, National Institute of Standards and Technology, and IEEE standards, thereby validating the accuracy and reliability of the proposed method for ultra-wideband measurements.

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I. INTRODUCTION

Amidst the technological trends of higher frequencies in electronic products, system integration, and wireless energy transmission, electromagnetic shielding has become the critical link connecting wireless energy innovation with the safe operation of electronic devices.^{1–4} Microwave wireless power transfer (MWPT) has expanded its application scope by virtue of high-efficiency antennas and rectification technologies. The limitations of antenna size and the radiative nature of energy have created an urgent demand for shielding to ensure personal safety. Functional complementarity between energy transmission systems and shielding materials can be achieved. This approach balances energy transmission efficiency with shielding effectiveness while adapting to diverse application scenarios, providing a core solution to the technical balancing challenge in this field.^{5,6} Currently, the rising

operating frequencies of electronic devices have propelled the 1–18 GHz band into mainstream use. Electromagnetic interference from high-frequency systems, such as 5G/6G and the Internet of Things, has become a significant industry challenge, making electromagnetic shielding materials a core solution. The integration of mobile terminals and the wideband interference challenges posed by electric vehicles further underscore the critical role of electromagnetic compatibility (EMC) in safeguarding electronic devices, thus making the development of highly effective shielding solutions a pressing priority.^{3,6–9} Meanwhile, breakthroughs in new shielding materials and the trend toward high-frequency equipment have made the establishment of a wideband shielding effectiveness testing system and the development of related technologies key research priorities.^{3,7} Therefore, the development and promotion of measurement standards for broadband electromagnetic shielding effectiveness are of significant theoretical and practical importance

for enhancing the safety of wireless power transmission systems and accelerating the engineering implementation of related technologies. Shielding effectiveness, as the core metric for evaluating a material's shielding performance, not only determines its electromagnetic characteristics but also serves as a crucial basis for guiding practical engineering applications.^{10–12} Accordingly, establishing a scientific and accurate method for measuring shielding effectiveness has notable theoretical value and practical significance for advancing electromagnetic shielding materials and enhancing the performance of electromagnetic protection systems.

As mentioned earlier, the increasing diversity of application scenarios is accelerating the evolution of materials' electromagnetic shielding performance toward wideband capabilities.¹³ It is worth noting that compared to applications in high-frequency bands, low-frequency electromagnetic shielding materials demonstrate greater technological value due to their unique electromagnetic wave attenuation mechanism. In strategic sectors, such as aerospace, new energy vehicles, and high-end medical equipment, low-frequency electromagnetic shielding materials have become critical factors in ensuring stable equipment operation and information security due to their exceptional performance.^{14,15} Meanwhile, measurement methods for the shielding effectiveness of electromagnetic shielding materials are continually optimized and improved to better meet research and application requirements. Among these, the development and refinement of broadband measurement technology have become a key direction in current research.^{15,16} Among various methods for measuring electromagnetic shielding effectiveness, the flange coaxial/coaxial method stands out as the most widely applied classic technique due to its significant advantages: a well-established principle, ease of operation, high precision, strong stability, and excellent repeatability. This method continues to play a pivotal role in electromagnetic shielding research. The characterization of shielding effectiveness based on this approach remains the industry-recognized mainstream technical pathway to date. However, its upper frequency limit is constrained to 1.5 GHz.^{17,18} In 2004, Tamburrano and Sarto upgraded the shielded test fixture to cover a frequency range from 30 MHz to 8 GHz. Still, the device remained constrained by its narrow bandwidth and low-frequency limitations.¹⁹ In 2005, Hatfield constructed an ultra-wideband test system covering 80 MHz–18 GHz, but its testing efficiency and accuracy still required improvement.²⁰ In 2006 and 2014, Sarto and Tamburrano successively developed and optimized broadband coaxial test fixtures spanning 30 MHz–18 GHz. While these designs reduced measurement errors, they failed to account for the effects of the coaxial cable's internal supporting medium and electromagnetic leakage/interference.^{7,21} In 2021, the Chen Chaohan research group measured thin materials from 30 MHz to 8 GHz using a coaxial flange-structure device. However, both the starting frequency and operational bandwidth still require further improvement.⁵ In 2022, Hyun Ho Park achieved shielding effectiveness measurements for thin materials within the 10 MHz–10 GHz range by increasing the inner and outer conductor radii of the coaxial flange. However, the high-frequency range remained limited to 10 GHz.⁷ Similarly, in 2025, Sankar *et al.* measured the shielding effectiveness of carbon fiber-nylon composites across the 100 MHz–8 GHz frequency range. However, the lowest measurement frequency still commenced at 100 MHz, failing to overcome the bottlenecks of low-frequency coverage and ultra-wideband measurement.⁹

Given that 10 MHz is the critical point at which the shielding mechanism transitions from being dominated by low-frequency magnetic fields to being dominated by high-frequency electric fields/plane waves, thin materials can be used to achieve efficient shielding at this frequency and above, avoiding the volume and mass redundancy issues caused by thick materials required in the low-frequency range.^{22–24} In addition, its wavelength is compatible with standard testing environments and equipment dimensions, covering common electromagnetic interference frequency bands to ensure the practicality and cross-platform comparability of test data. Based on the aforementioned technical and standard advantages, 10 MHz has been widely adopted as the universal minimum frequency for electromagnetic shielding effectiveness testing.^{25,26} However, existing electromagnetic shielding effectiveness measurement methods are generally constrained by narrow operating frequency bands, making it difficult to achieve full-band coverage across the ultra-wideband testing range from 10 MHz at low frequencies to 18 GHz at high frequencies. To overcome this technical limitation, this paper achieved a synergistic optimization of low-frequency extension and full-band coverage by specifically optimizing the design of the supporting medium and key configuration parameters of the flange. This not only successfully extended the measurement lower limit to 10 MHz but also established an ultra-wideband measurement system spanning 10 MHz–18 GHz. Ultimately, it provides a reliable and efficient technical solution for ultra-wideband electromagnetic shielding effectiveness measurements. Carbon nanotubes were used as representative verification samples to compare the performance of the self-developed coaxial flange transmission line structure with that of the X/Ku-band waveguide structure specified in standard methods. Experimental data demonstrate that within the 10 MHz–18 GHz ultra-wideband range, the measurement deviation between the two configurations is stably constrained to less than 0.5%. This quantitative indicator fully verifies the exceptional accuracy and reliability of the proposed improved method for electromagnetic shielding effectiveness measurements.

II. THEORY AND METHODS

The structure of the coaxial transmission line used for shielding effectiveness testing is shown in Fig. 1.

The coaxial transmission line is designed with an inner conductor outer diameter of 3.04 mm and an outer conductor inner

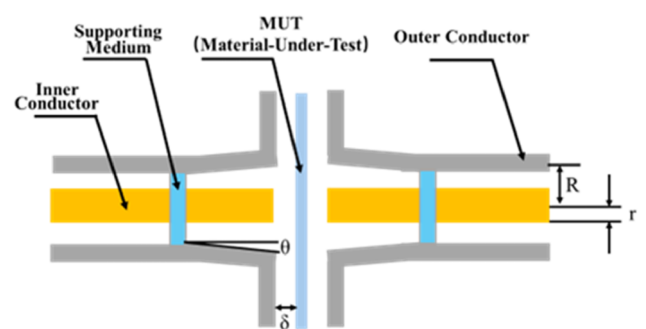


FIG. 1. Schematic diagram of coaxial transmission line structure.

diameter of 7 mm, covering frequencies from DC to 18 GHz. A supporting medium must be inserted between the inner and outer conductors to ensure coaxiality and perpendicularity. A flange structure is employed to guarantee electromagnetic leakage suppression. The overall design ensures low loss and low reflection across the entire measurement standard throughout the full frequency range.

A. Design theory of coaxial flanges

1. Electromagnetic field analysis of coaxial flanges

For flanges employed in coaxial line-based measurements of material shielding effectiveness, the core technical merit of flange design lies primarily in the following three aspects:²⁷

- (I) Mechanical connection and electromagnetic leakage protection fixed flange: Achieves rigid mechanical connection between components of the coaxial system, ensuring overall structural stability while facilitating convenient disassembly and reassembly of the assembly.
- (II) Electrical continuity and impedance-matched flanges ensure continuous electrical connection between the inner and outer conductors of coaxial systems, guaranteeing low reflection and low loss.
- (III) Through precise design of key parameters, such as flange surface flatness and contact component dimensions, impedance matching can be achieved within the system, significantly reducing electromagnetic field leakage.

When a coaxial transmission line connects to a flange, the electromagnetic field distribution changes due to structural discontinuities. The following analysis examines typical scenarios.

a. Electric field concentration at flange gaps. Due to the edge effect between flanges, if a minute gap ($\delta \leq 0.1$ mm) exists on the flange contact surface, an electric field peak $E_{\max}$ forms at the gap edge. Its distribution can be approximated using conformal mapping,

$$E_{\max} \approx E_0 \left(1 + \frac{\pi R}{2\delta} \right), \quad (1)$$

where E_0 is the ideal coaxial electric field strength, R is the outer radius, and δ is the gap width. $E_{\max}$ increases as δ decreases, with $E_{\max} \rightarrow E_0$, indicating that good contact can suppress electric field concentration.

In practical engineering applications, abrupt dimensional changes at flange interfaces typically cause electric field concentration, resulting in localized high-field-strength regions.

b. Magnetic field disturbance in the flange transition section. Flange connections cause the current path to deviate from the ideal circumferential distribution, generating an axial magnetic field component H_z whose intensity correlates with the geometric parameters of the transition section. For tapered transition flanges (taper angle θ), the axial magnetic field can be approximated as

$$H_z \approx \frac{I \cdot \theta}{4\pi d}. \quad (2)$$

Here, θ denotes the cone angle and d represents the vertical distance from the field point to the central axis. When the cone angle

$\theta \rightarrow 0$, $H_z \rightarrow 0$, indicating that smooth transitions can reduce magnetic field disturbances.

c. Coaxial flange excitation of higher-order TE/TM modes. Flange discontinuities excite higher-order TE/TM modes, whose cutoff frequencies are dependent on structural dimensions as follows: Cutoff frequencies for TE and TM modes: For radial dimensional discontinuities caused by coaxial flanges (e.g., outer conductor inner diameter changing from D_1 to D_2 , inner conductor outer diameter changing from d_1 to d_2), the cutoff frequencies for higher-order TE modes are approximately

$$f_{c,TE_{mn}} = \frac{c}{2\sqrt{\mu\epsilon}} \cdot \frac{\lambda_{mn}}{D_{\text{avg}}}, \quad (3)$$

where c is the speed of light in vacuum and μ and ϵ are the magnetic permeability and dielectric constant of the medium. D_{avg} is the average diameter at the flange,

$$D_{\text{avg}} \approx \frac{D_2 + d_2}{2}. \quad (4)$$

Here, D and d denote the diameters of the outer and inner conductors, i.e., $2R$ and $2r$. λ_{mn} is the mode constant, determined by the roots of the derivative of the Bessel function.

The difference between the TE mode and TM mode lies in the determination of the roots of the Bessel function; the TM mode corresponds to the n th root of the first derivative of the Bessel function. The n th root of the first-order derivative of the Bessel function corresponds to the TM mode. The n th root of the Bessel function corresponds to the TE mode.

2. Flange design

The design elaborated herein integrates flanges at the ports of the coaxial transmission line, with interconnections established through flange interfaces. In this transmission line setup, the flanges serve as key components for joining two coaxial cable segments. Compared with the direct connection schemes specified in ASTM standards and the relevant literature, these flanges not only improve the line's transient transmission performance but also ensure connection reliability and immunity to interference. Furthermore, they effectively enhance the system's anti-jamming capability and signal confidentiality. According to IEC 60339-2:1996, the flange outer diameter is calculated using formula (5),²⁸

$$R_{\text{flange}} = r_{\text{outer}} + L_{\text{pad}} + L_{\text{bolt}} + C, \quad (5)$$

where r_{outer} is the inner radius of the outer conductor, L_{pad} is the width of the conductive gasket mounting groove (typically 2–5 mm), L_{bolt} is the radius of the bolt head, C is the edge allowance (typically 3–8 mm, ensuring flange structural strength).

The flange thickness is determined by formula (6),

$$T \geq \frac{c}{20f_{\max}}. \quad (6)$$

Based on the theoretical support provided by Eqs. (1)–(6), the flange material is made identical to that of the coaxial transmission line to maximize flange performance. Subsequently, optimization is performed on the geometric dimensions of the flange, with its specific design parameters presented in Table I.

TABLE I. Optimized parameters of the flange.

No.	Category	Measurement error
1	R_{flange}	17 mm
3	r_{out}	7 mm
4	L_{pad}	3 mm
5	L_{bolt}	4 mm
6	C	5 mm
7	f_{max}	20 GHz
8	T	3 mm

B. Design theory of support medium

The introduction of a support medium will degrade the broadband and transmission characteristics of the coaxial cable. Therefore, the geometric shape, material selection, inner and outer diameter dimensions, and installation position of the support medium are critical factors that determine the overall performance of this transmission line.

1. Structural design of the support medium

Regarding the geometric configuration of the supporting medium, previous studies have been conducted. This paper adopts a six-hole structure to minimize the impact of the supporting medium on the coaxial transmission line's performance, as shown in Fig. 2.

The core concept of this configuration is to minimize the equivalent dielectric constant while ensuring the necessary mechanical strength, thereby reducing the disturbance of the supporting medium to the electromagnetic properties to the lowest possible level.

According to the theory of volume reduction optimization for dielectric materials, an excessive volume fraction of solid support

media can interfere with electromagnetic field distribution by accumulating internal polarization charges, thereby exacerbating signal loss and leakage.^{29,30} The six-hole structure significantly reduces the volume of solid dielectric material by incorporating six through-holes, effectively minimizing polarization interference. Simultaneously, the number of holes achieves a precise balance between "volume reduction" and "structural support"—too few holes result in insufficient polarization suppression due to excessive residual dielectric material, while too many holes excessively weaken the dielectric structure's strength and increase susceptibility to new leakage points caused by manufacturing defects. This design both mitigates polarization interference through rational volume reduction and ensures reliable support by leveraging a stable six-hole structure, ultimately achieving synergistic optimization of dielectric properties and electromagnetic suppression.

Selecting a six-hole structure for the coaxial flange supports medium balances practical requirements, including load-bearing stability, ease of assembly and maintenance, electromagnetic compatibility, and heat dissipation. It represents one of the optimal solutions that balances theoretical adaptability and engineering practicality.

2. Selection of materials for the support medium

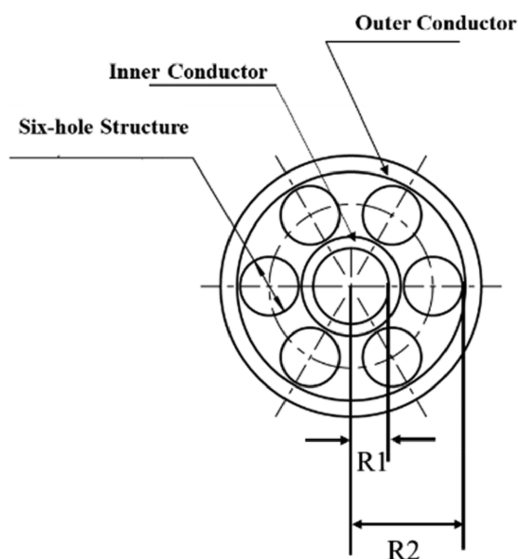
When introducing a dielectric support ring into a coaxial transmission line, maintaining a 50 Ω characteristic impedance match requires that the support material's electromagnetic properties closely resemble those of air. This minimizes electromagnetic wave reflection at the air–dielectric interface, thereby reducing interference with broadband transmission performance. At the same time, the medium must also possess sufficient mechanical strength and thermal stability to ensure long-term maintenance of coaxiality and perpendicularity, while meeting positioning and reliability requirements in complex environments.

This paper selects polyetheretherketone (PEEK) for its superior electromagnetic and mechanical properties: lighter than metal, resistant to high temperatures, high strength, and chemically resistant, and it can directly replace metal in harsh environments. Compared to traditional engineering plastics, it offers comprehensive advantages in both design and functionality.

3. Dimension parameters of the support medium

Inner and outer diameters of the support medium: Since the support medium must ensure axial alignment of the inner and outer conductors, its inner radius must be smaller than the inner conductor's radius. In comparison, its outer radius must be larger than the outer conductor's radius. This allows the support medium to be clamped between the inner and outer conductors, enabling it to perform its supporting function. The insertion of the support medium between the inner and outer conductors inevitably affects the impedance of the coaxial line. Therefore, the inner radius (R_1) and outer radius (R_2) of the support medium become particularly critical.

As documented, the electromagnetic parameters and thickness of the supporting medium also exhibit a certain functional relationship. If non-magnetic materials are used as the medium, the following formula applies:³¹

**FIG. 2.** Structure dimensions structure of the support medium.

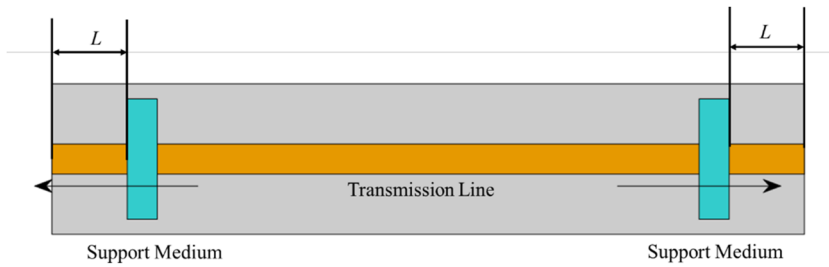


FIG. 3. Schematic diagram of supporting medium position.

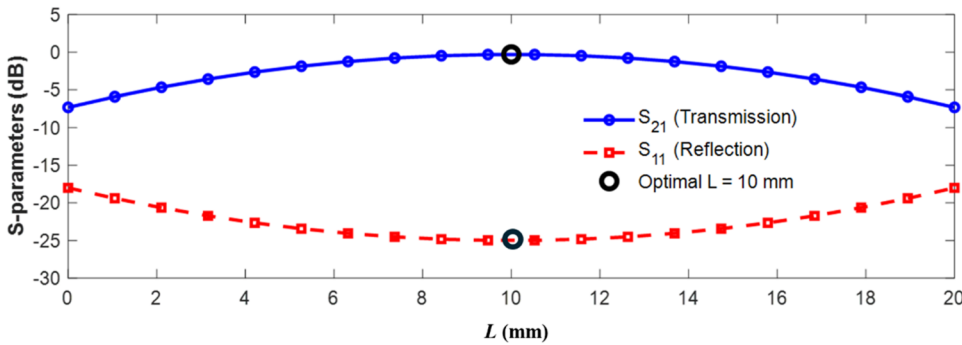


FIG. 4. Simulation results for supporting medium position.

$$B = \frac{\lambda_g}{\pi} \tan^{-1} \sqrt{\frac{\left(\frac{f_c}{f}\right)^2 - 1}{\epsilon_r - \left(\frac{f_c}{f}\right)^2}}, \quad (7)$$

where B represents the thickness of the insulating support medium, λ_g denotes the waveguide wavelength within the medium, ϵ_r is the dielectric constant of the supporting medium, f_c indicates the cutoff frequency for higher modes of the coaxial line, and f signifies the maximum operating frequency of the coaxial line.³⁰

The above formula applies to all TE modes. From the formula, it follows that when the thickness remains constant, the resonant frequency decreases as the relative permittivity of the insulating support medium increases. When using the same material as the insulating support medium, the operating frequency can be progressively increased as the thickness decreases. When the thickness reduces to zero, the structure becomes a complete coaxial airline, and no higher-order modes are excited. When the thickness of the insulating support medium approaches the inner diameter R_2 of the outer conductor, the transmission of electromagnetic waves within the coaxial cable becomes highly unstable due to the resonant effect of the insulating support medium. This causes a corresponding decrease in the cutoff frequency of the coaxial cable. Therefore, the thickness of the insulating support medium should be moderate. Excessive thickness readily causes resonance instability, whereas appropriate thinning increases the operating frequency and suppresses higher-order modes. In addition, given the proximity effect, it is essential to ensure that the thickness of the insulation support medium exceeds the depth of the slots in the outer conductor. In addition to the above factors, it is also necessary to ensure that the thickness of the insulating support medium provides proper positioning between the inner and outer conductors and their coaxiality.²⁹

The schematic diagram illustrates that the distance between the supporting medium and the port is presented in Fig. 3. Traditional approaches typically focus on the spacing between medium plates. In this work, only the distance L from the medium plate to the port needs to be determined.

Numerical simulations with different values of distance L were carried out using the finite element method (FEM), with the results illustrated in Fig. 4. Based on the simulation results, the optimal distance from the port is 10 mm.

C. Analysis of operating frequency for coaxial flange structures

1. Theoretical analysis of maximum frequency

The maximum operating frequency of a coaxial transmission line is primarily determined by its mode cutoff frequency. The core objective is to suppress higher-order modes (TE_{11}) during transmission, ensuring that only the fundamental (TEM) mode propagates. Calculating this cutoff frequency requires integrating the transmission line's structural dimensions with the characteristics of its dielectric medium.^{27,28} The cutoff frequency is calculated as shown in (8),

$$f_{c(TE_{11})} = \frac{c}{2\pi\sqrt{\mu_r\epsilon_r}(r+R)}, \quad (8)$$

where c is the speed of light in vacuum, μ_r is the magnetic permeability of the medium, ϵ_r is the relative permittivity of the medium, a is the outer radius of the inner conductor, and b is the inner radius of the outer conductor.

This measurement structure employs a 7 mm coaxial configuration ($a = 3.04$ mm; $b = 7$ mm) with air as the dielectric. Manufactured and designed in strict accordance with the formula, the calculated

cutoff frequency $f_{c(TE_{11})} = 19.4$ GHz ensures coverage up to 18 GHz.

2. Theoretical analysis of minimum frequency

To extend the measurable minimum frequency of coaxial flange transmission lines, the core approach involves reducing the transmission line's inherent losses, optimizing impedance matching to minimize low-frequency signal reflections, and adapting to the propagation characteristics of low-frequency electromagnetic fields. This approach is underpinned by transmission line theory, low-frequency electromagnetic loss theory, and impedance matching theory.

a. Transmission line theory. At low frequencies (where wavelength λ is much greater than transmission line length l , i.e., $l \ll \lambda/20$), transmission lines exhibit lumped parameter characteristics. Signal attenuation is primarily due to conductor ohmic losses and dielectric losses in the medium, rather than to skin-effect losses characteristic of high frequencies.³⁰

To maintain the standard 50 Ω characteristic impedance in the design, the diameters of the inner and outer conductors were increased (to 7 mm in this design) to reduce the conductor's distributed resistance. Simultaneously, a low-dielectric material was selected to prevent impedance deviation as frequency decreases. Control transmission line length $l \ll \lambda_{\min}/20$ (where $\lambda_{\min}$ is the wavelength corresponding to the lowest frequency) to ensure that low-frequency signals remain in a quasi-static transmission state, thereby reducing waveguide mode conversion and standing wave interference.

The length of the coaxial flange transmission line in this paper is calculated assuming a minimum frequency of 10 MHz, corresponding to a wavelength of 30 m. The length is set to 20 mm, which is significantly less than one-twentieth of the wavelength.

b. Theory of low-frequency electromagnetic losses. For coaxial flange transmission lines, electromagnetic losses in the low-frequency range are primarily conductor losses and dielectric losses, described, respectively, as follows:³⁰

(a) Conductor losses.

At low frequencies, the skin depth increases [as shown in formula (9)],

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}. \quad (9)$$

Here, ω , μ , and σ are angular frequency, permeability, and conductivity, respectively.

Electric current can penetrate the interior of a conductor, with losses determined by the conductor's total resistance. The formula for resistance per unit length is shown in (10),

$$R = \frac{1}{\sigma} \left(\frac{1}{\pi d \delta} + \frac{1}{\pi D \delta} \right). \quad (10)$$

Here, d and D denote the inner and outer conductor diameters, respectively. Higher conductivity and larger conductor cross-sectional area result in lower losses.

(b) Dielectric loss.

The dielectric loss is characterized by the tangent of the loss angle. The dielectric loss constant per unit length is given by Eq. (11),

$$\alpha_d = \frac{\pi f \epsilon_0 \epsilon_r \tan \delta}{c \sqrt{\epsilon_r}}, \quad (11)$$

where f is the operating frequency, c is the speed of light in vacuum, and $\tan \delta$ denotes the dielectric loss tangent.

In summary, the minimum frequency can be expressed using the following formula (12):

$$f_{\min} = \frac{c}{2\pi(D+d)\sqrt{\epsilon_r\mu_r}} \cdot \frac{1}{1 - \frac{\alpha_c}{\alpha_{c0}} - \frac{\alpha_d}{\alpha_{c0}}}. \quad (12)$$

Here, α_c represents conductor loss, α_d represents dielectric loss, and α_{c0} is the reference cutoff attenuation constant for the TE_{11} higher-order mode of a coaxial cable in a lossless air dielectric, which is determined solely by the structural dimensions.

Although the frequency f decreases at low frequencies, the polarization relaxation effect of the dielectric may cause $\tan \delta$ to increase, necessitating the selection of low-loss dielectric materials.

This design employs high-conductivity materials—oxygen-free copper ($\sigma = 5.8 \times 10^7$ S/m) and silver-plated copper ($\sigma = 6.3 \times 10^7$ S/m, with silver-plated copper exhibiting higher conductivity than gold-plated copper)—for the conductors. The cross-sectional area of both inner and outer conductors is increased to reduce conductor losses. The hybrid structure combining “air” and a “supporting medium” reduces medium loss, further ensuring low-frequency operation at 10 MHz.

III. DESIGN AND EXPERIMENTATION

A. Design and processing

As derived from the theoretical analysis in the prior section, the three-dimensional structural model of the developed measurement standard is presented in Fig. 5.

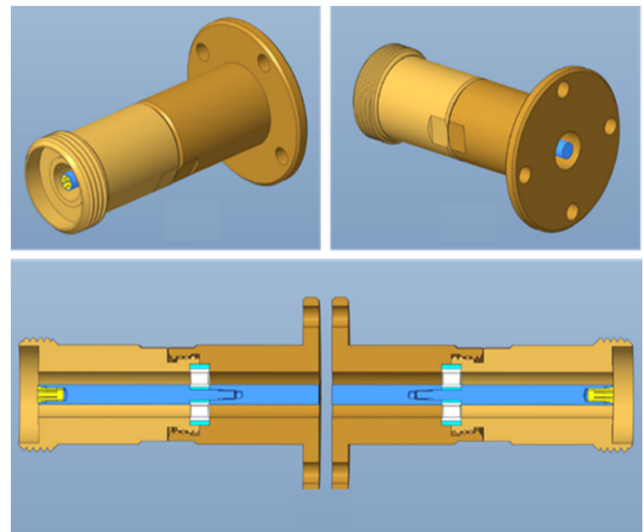


FIG. 5. Three-dimensional structural model for shielding the measurement flange coaxial assembly.



FIG. 6. Device for shielding the coaxial structure of a measurement flange.

Based on the model and engineering drawings, subsequent machining and assembly procedures may be carried out. Care should be taken to ensure that the end face of the inner conductor is recessed by 0–0.01 mm relative to that of the housing, to avoid deformation of the inner conductor caused by excessive protrusion during assembly. The inner conductor end face of the test end must be 0–0.01 mm higher than the housing end face to ensure full contact between the inner and outer conductors of the mating material. The machined structure and physical photograph of the fabricated transmission line are presented in Fig. 6. The upper-left inset shows the top view and details the support components, inner conductor, outer conductor, and flange structure of the fixture. The upper-right image provides a side view, and the lower inset demonstrates the assembled state of the interconnected components.

The insertion loss and standing-wave ratio of the fabricated coaxial shielded transmission line measurement setup were characterized using a vector network analyzer, and the corresponding results are presented in Fig. 7. The results demonstrate that the coaxial flange structure designed based on theoretical calculations exhibits a measured insertion loss of 0.1 dB and a maximum VSWR of 1.025. These performance metrics significantly exceed the general standard requirements for RF transmission devices, which specify insertion loss ≤ 0.5 dB and VSWR ≤ 1.20 , fully highlighting the structure's remarkable superiority in signal transmission performance.^{31,32}

B. Experiment and validation

1. Experimental system and verification materials

To verify the effectiveness of the method proposed in this paper, the shielding effectiveness of carbon nanotubes—a widely studied advanced material—was measured using an R&S ZVA40 vector network analyzer and the custom-developed transmission line measurement setup reported herein. The left inset in Fig. 8 shows the testing system, and the right inset presents the sample mounting fixture. The black sheet serves as the material under test (MUT). The test system shown in Fig. 8 verified the electromagnetic shielding performance measurements of carbon nanotube materials using the coaxial flange structure device described in this chapter across the 0.01–18 GHz frequency range, as well as using a metal rectangular waveguide in the X and Ku bands.

2. Comparison and verification results

As shown in Fig. 8, using the coaxial flange structure proposed in this chapter, the electromagnetic shielding performance of carbon nanotube samples was measured over the frequency range of 0.01–18 GHz, and the results were validated against those obtained from metallic rectangular waveguides in the X-band and Ku-band. The waveguide method employed in this study is valid only for

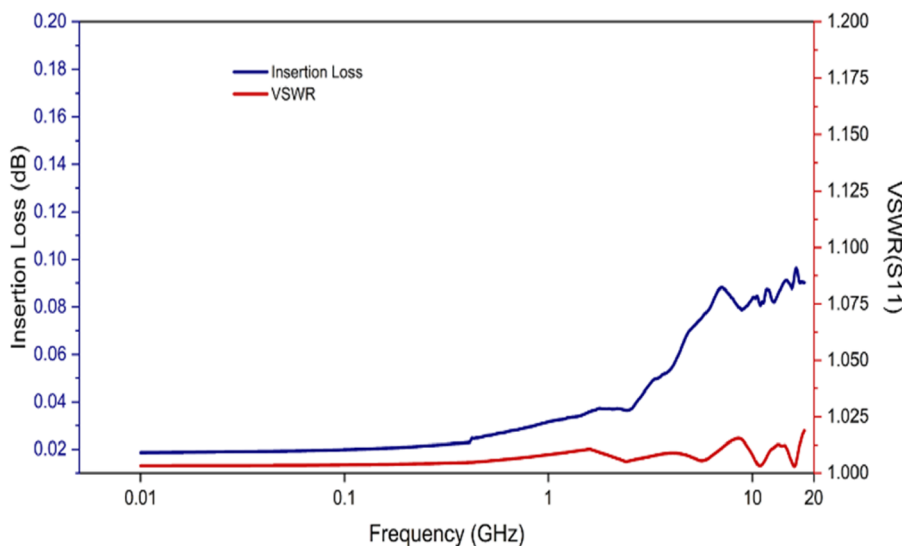


FIG. 7. Insertion loss and VSWR of coaxial flange-based measurement devices.

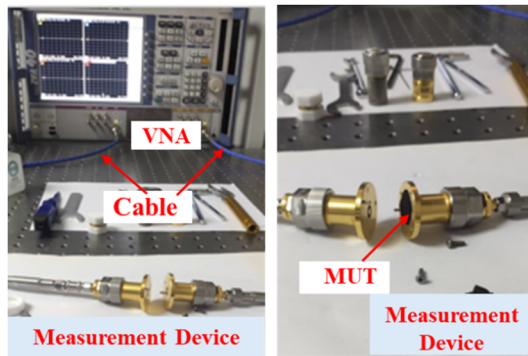


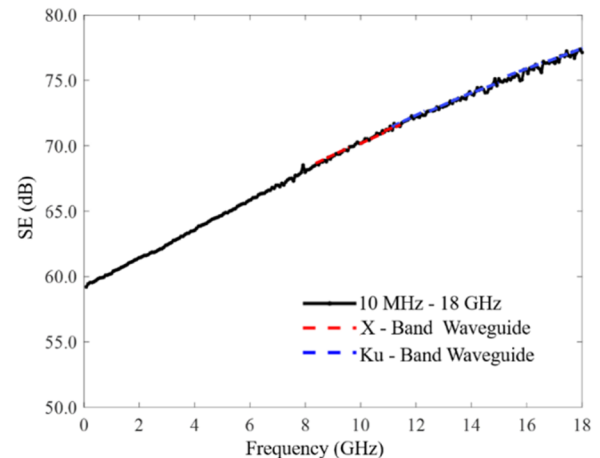
FIG. 8. Material shielding effectiveness measurement devices.

frequencies above the waveguide's cutoff frequency. Extending the measurement range to lower frequencies (e.g., 10 MHz) is constrained by the excessively large physical size of waveguides at low frequencies, necessitating alternative measurement setups, such as coaxial lines. This extended work will be addressed in future studies. To facilitate direct comparison among the three figures, a unified axis scale was employed, and the results are presented in Fig. 9.

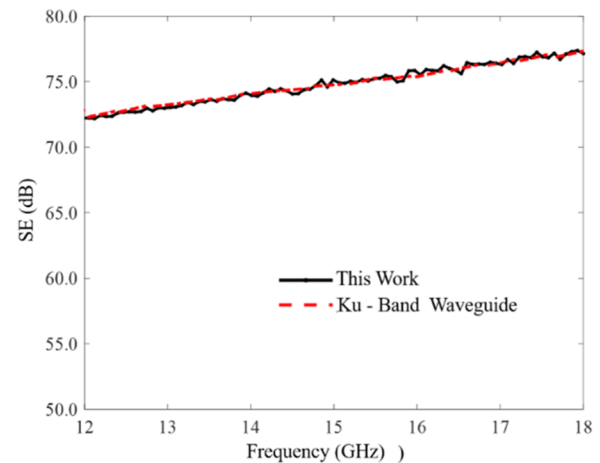
The solid black line in the figure represents the measurement results of the transmission line structure device designed and developed in this paper. In contrast, the dashed red lines represent the measurement results of the X-band and Ku-band rectangular waveguides, respectively. Generally, rectangular waveguides serve as a benchmark for validating coaxial transmission line measurements.^{8,13} The calculated errors and uncertainties corresponding to the X-band and Ku-band are presented in Table II based on Figs. 9(a) and 9(b).

The analysis of the measured results shows that the average deviation between the proposed method and the rectangular waveguide measurement is 0.50%, with an associated uncertainty of 0.17%. The relevant parameters were converted with reference to a shielding effectiveness of 50 dB. Compared with similar measurement devices described in the existing literature, the results derived from this study exhibit the following advantages, as presented in Table III.

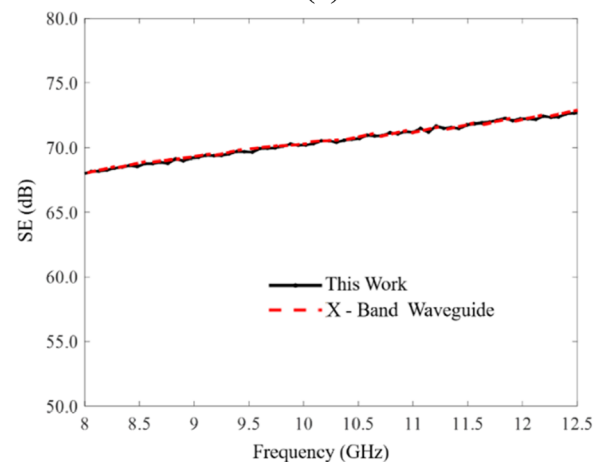
Per relevant standards from ASTM, NIST, and the Institute of Electrical and Electronics Engineers (IEEE), a 50 dB electromagnetic shielding effectiveness corresponds to a 99.999% attenuation efficiency for electromagnetic waves. This value serves as a benchmark metric for evaluating electromagnetic shielding performance.^{33–35} Based on the Guide to the Expression of Uncertainty in Measurement (GUM) methodology specified by the Joint Committee for Guides in Metrology (JCGM) and the Bureau International des Poids et Mesures (BIPM), the allowable measurement uncertainty for this benchmark at a 99% confidence level is $\leq 0.515\%$.^{36–38} However, in accordance with GUM guidelines, the uncertainty derived from Table II and the maximum error-induced uncertainty in Table III yield a combined uncertainty of 0.49%. Therefore, the findings of this study align with industry requirements. This demonstrates that the methodology proposed in this chapter is both scientifically sound and effective.



(a)



(b)



(c)

FIG. 9. Comparison of measurement results for thin materials: this device vs X/Ku-band waveguides. (a) Comparison with X-band waveguide measurements; (b) comparison with Ku-band waveguide measurements; and (c) combined comparison with standard X- and Ku-band waveguide measurements.

TABLE II. Analysis of errors and uncertainties in the measurement of thin materials.

Frequency	SE		Error (%)
	Standard waveguide	This work	
8.2 GHz	69.1566 dB	68.7677 dB	0.56
9 GHz	70.2195 dB	69.8509 dB	0.52
10 GHz	71.2729 dB	70.8966 dB	0.53
11 GHz	72.0260 dB	71.7830 dB	0.34
12 GHz	72.7741 dB	72.4131 dB	0.50
13 GHz	73.2265 dB	73.0249 dB	0.28
14 GHz	74.1473 dB	73.9449 dB	0.27
15 GHz	74.8876 dB	75.1229 dB	0.31
16 GHz	75.9587 dB	75.3410 dB	0.81
17 GHz	76.8074 dB	76.3038 dB	0.66
18 GHz	77.8856 dB	77.3597 dB	0.68
Mean error		0.50%	
Maximum error		0.81%	
Uncertainty		0.17%	

TABLE III. Literature-reported performance comparison of similar fixtures for SE measurement. Boldface denotes the optimal values.

No.	Frequency range	Measurement error	Reference
1	30 MHz–1.5 GHz	±3 dB	17 and 18
2	100 MHz–8 GHz	N/A	9
3	30 MHz–8 GHz	N/A	19
4	10 MHz–10 GHz	N/A	7
5	80 MHz–18 GHz	N/A	20
6	30 MHz–18 GHz	N/A	22
7	10 MHz–18 GHz	±1.9 dB	This work

IV. CONCLUSION

To meet the demand for accurate and reliable measurement of electromagnetic shielding effectiveness (EMSE) of thin planar materials over the ultra-wide frequency band spanning 10 MHz–18 GHz, this study proposes targeted improvements to the coaxial flange transmission line measurement method. Building on this optimized methodology, a custom-built coaxial shielding measurement apparatus was successfully fabricated, which enables high-precision characterization of EMSE for thin materials across the entire 10 MHz–18 GHz frequency range. Using carbon nanotube materials as the representative validation specimen, the test results obtained with the proposed method were compared with measured data from conventional standard waveguide measurement systems. The comparative results demonstrate that the measurement deviation between the two approaches is constrained to within 0.5%, fully satisfying the relevant requirements specified in ASTM, NIST, and IEEE standards.^{36–38} This result fully validates the scientific rigor and engineering practicality of the improved measurement method proposed in this paper, providing reliable technical support for testing the shielding effectiveness of thin materials across a wide frequency range, particularly in the low-frequency band.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Linggang Meng: Data curation (equal); Formal analysis (equal); Writing – original draft (equal). **Changjun Liu:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal). **Zhongqi He:** Investigation (equal); Resources (equal); Software (equal). **Shan He:** Investigation (equal); Resources (equal); Software (equal); Supervision (equal). **Chongwei Liao:** Supervision (equal); Validation (equal); Visualization (equal). **Zhifeng Zhang:** Validation (equal); Visualization (equal).

DATA AVAILABILITY

All data generated and analysed in this work are included within this article.

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